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Evaluation of Tree-Based Voting Algorithms in Water Quality Classification Prediction

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Abstract: Accurately predicting the state of surface water quality is crucial for ensuring the sustainable use of water resources and environmental protection. This often requires a focus on the range of factors affecting water quality, such as physical and chemical parameters. Tree models, with their flexible tree-like structure and strong capability for partitioning and selecting influential water quality features, offer clear decision-making rules, making them suitable for this task. However, an individual decision tree model has limitations and cannot fully capture the complex relationships between all influencing parameters and water quality. Therefore, this study proposes a method combining ensemble tree models with voting algorithms to predict water quality classification. This study was conducted using five surface water monitoring sites in Qingdao, representing a portion of many municipal water environment monitoring stations in China, employing a single-factor determination method with stringent surface water standards. The soft voting algorithm achieved the highest accuracy of 99.91%, and the model addressed the imbalance in original water quality categories, reaching a Matthews Correlation Coefficient (MCC) of 99.88%. In contrast, conventional machine learning algorithms, such as logistic regression and K-nearest neighbors, achieved lower accuracies of 75.90% and 91.33%, respectively. Additionally, the model's supervision of misclassified data demonstrated its good learning of water quality determination rules. The trained model was also transferred directly to predict water quality at 13 monitoring stations in Beijing, where it performed robustly, achieving an ensemble hard voting accuracy of 97.73% and an MCC of 96.81%. In many countries' water environment systems, different water qualities correspond to different uses, and the magnitude of influencing parameters is directly related to water quality categories; critical parameters can even directly determine the quality category. Tree models are highly capable of handling nonlinear relationships and selecting important water quality features, allowing them to identify and exploit interactions between water quality parameters, which is especially important when multiple parameters together determine the water quality category. Therefore, there is significant motivation to develop tree model-based water quality prediction models.

Keywords: water quality; machine learning; ensemble learning; voting algorithm; tree model

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1. Introduction

Water resources constitute an indispensable component of human society and natural ecosystems, covering approximately 71% of the Earth's surface. However, the freshwater available for drinking, agriculture, and industrial use is extremely limited, accounting for only about 2.5% of the total water resources. With the increase in the global population and the evolution of societal processes, the demand for high-quality water is urgent. Historically, nearly every country in the world has experienced short-term water shortages, with more than 80 countries facing severe shortages [1]. Water resources are influenced by various factors, with natural and anthropogenic factors being the main determinants of water quality parameters [2]. The impact of land use and seasonal factors on water quality varies with the type of pollutants [3], and considering the interaction between land use and climate change, their combined impact on the hydrological environment

of watersheds is more obvious and significant than the impact of land use alone [4]. Sustainable utilization of water resources is a global and complex task, and different categories of water quality have different uses, such as drinking water, agricultural water, and industrial water, making it an important factor affecting the efficiency of water resource use [5]. Eutrophication of water bodies is a significant environmental issue worldwide. The level of eutrophication is influenced by human activities [6], and certain water quality parameters, such as phosphorus and nitrogen concentrations, need to be focused on [7,8]. The concentration of dissolved oxygen significantly influences the aquatic environments such as rivers [9] and ponds [10], having a direct correlation with the quality of water. The concentrations of total phosphorus [11,12] and total nitrogen [12] serve as indicators of the eutrophication level in water bodies, profoundly affecting water quality and the activity of aquatic life. The concentrated distribution of pollution sources and adverse hydrodynamic conditions exacerbate eutrophication [13], while organic matter in sewage also damages river ecosystems [8]. Protecting existing water resources and taking preventive measures against pollution are necessary conditions for better utilizing water resources.

Therefore, it is necessary to evaluate water quality based on physical, chemical, and microbial parameters. The Water Quality Index (WQI) is such a comprehensive assessment tool, helpful for quickly understanding the state of water quality. It normally requires the input of a set of representative parameters and assigning a standardized score to these parameters [14], determining their weights based on their importance to water quality, and finally calculating the WQI [15]. When calculating the WQI, the weights are not constant and can differ based on different evaluation systems and purposes [16]. Of course, water quality categories can also be divided based on the WQI value, thus intuitively reflecting the state of water quality [17,18].

Securing an abundant and high-quality surface water supply is a forward-looking challenge. Timely and accurate water quality predictions can alert to potential pollution crises [19], especially in situations where water bodies are eutrophic and water quality parameters are uncertain [20]. The quality of water is related to the sustainable development of human society. With scientific information technology, such as sensor technology [21], big data analysis and mining of water quality [21], and artificial intelligence and machine learning methods [22], complex calculations can be efficiently carried out, which is conducive to improving the level of monitoring and ensuring water quality. Programs designed using dynamical knowledge and coupled systems can detect changes in certain chemical parameters, such as phosphorus concentration dynamics [23], and timely satellite information can also reveal future trends of water quality deterioration and eutrophication levels [24]. Remote sensing data complement sensor data to estimate abnormal conditions of water quality parameters, thereby assessing pollution conditions, such as using anomalies in monitoring water temperature to estimate thermal pollution [25]. Simultaneously, the future variations in water quality indicators are critical areas of focus. A study proposed a model based on a dual-attention mechanism that considered both time and dimension, achieving over 80% accuracy in out-of-sample predictions [26]. In recent years, machine learning models have gained increasing attention for water quality prediction. The use of artificial neural networks (ANNs) for simulating groundwater indices has achieved an R-squared (R^2) value of 0.9984, reaching an ideal accuracy level [27]. Another study has devised a feedforward three-layer perceptron neural network model capable of explaining approximately 95.4% of the variations in actual WQI and has simplified the traditional WQI computation [28]. However, these studies did not compare the performance of multiple machine learning models.

The employment of support vector machines (SVMs), neural networks (NNs), deep neural networks (DNNs), and K-nearest neighbors (KNNs) for real-time water quality monitoring has found the DNNs model to have the best classification accuracy, reaching 93% [29]. For groundwater quality prediction, random forest (RF), extreme gradient boosting (XGBoost), ANNs, and a deep learning model have been used, with the deep learning model achieving the highest performance, an R^2 value of 0.996. XGBoost has

exhibited superior predictive performance compared to RF and ANNs models with an R^2 value of 0.927 [30]. Cascade forward neural networks and radial basis function neural networks have been employed to predict temporal changes in river water quality, with the cascade forward neural network achieving more accurate short-term predictions [16]. The integration of the Internet of Things, deep learning technologies, and cloud platforms enabled the simple recurrent unit (SRU) model to more effectively predict water quality, thereby facilitating intelligent management of aquaculture [31]. A study has utilized DNNs, gradient boosting machine (GBM), and XGBoost, three mainstream machine learning models, to predict two key water quality indicators, showing that DNNs has outperformed the others with the lowest root mean square error (RMSE) of only 0.037 [32]. For lake water quality prediction, an improved decision tree algorithm has demonstrated a higher recognition rate and prediction accuracy than ANNs, and compared to the conventional C4.5 decision tree algorithm, it has had a more concise structure [33]. Logistic regression (LR), SVMs, KNNs, RF, and GBM have been employed for predicting and classifying the Water Quality Index, with the LR model achieving the highest accuracy of 90% [34]. Additionally, considering the spatial characteristics of multidimensional data, a fusion spatio-temporal graph convolution neural network (FSGCN) was utilized for spatio-temporal water quality prediction, outperforming other methods such as ANNs [35].

Aside from singular model strategies, ensemble machine learning models, showcasing the ensemble idea, have been widely used in water quality prediction and have exhibited good performance. A hybrid deep learning model integrating convolutional neural networks (CNNs), long short-term memory (LSTM), and attention mechanisms proved feasible for multi-step medium-term predictions [36]. A study introduced a deep learning-based ensemble method, wherein the STNX model (with spatio-temporal attention) demonstrated the highest performance for both short-step and long-step water quality predictions [37]. The RF model has been utilized for predicting the regional distribution of groundwater quality, demonstrating RF's excellent capability to learn nonlinear relationships between variables and identify key variables [38]. Key indicators reflecting groundwater quality hardness have been researched using boosted regression trees (BRT) and RF, effectively predicting the sensitivity to groundwater hardness [39]. Four machine learning models, utilizing ensembles of weak learners, have been employed to predict water quality classification. Among them, the GBM has exhibited the best predictive performance, achieving an accuracy of 99.5% [40].

Such combinations of weak learners also displayed their advantages. Another study combined CNNs and recurrent neural networks (RNNs) to predict pH levels and total nitrogen content, achieving superior prediction quality compared to other conventional deep learning models [41]. Six models, including XGBoost, RF, SVMs, ANNs, KNNs, and a gaussian classifier model (GCM), have been used for evaluating and predicting groundwater quality indicators, with RF being considered the best model due to its highest accuracy [15]. In predicting watershed water quality, the RF model has demonstrated higher predictive accuracy compared to the sequential minimal optimization support vector machine (SMO-SVM) [42]. Moreover, an ensemble of eight machine learning models has shown outstanding performance in predicting WQI, especially decision trees, extra trees, and XGBoost models, achieving perfect RMSE and R^2 values of 0.0 and 1.0 [43]. An ensemble of adaptive boosting (AdaBoost), RF, SVMs, and GBM has been used for classifying water quality states, with the GBM method achieving an accuracy of up to 100% [44]. Research incorporating explainable artificial intelligence (XAI) techniques reported that XGBoost achieved an R-squared value of 0.992 in water quality prediction [45].

Further practice of ensemble learning concepts involved creating more flexible strategy combinations. A study has applied seven machine learning classifiers for water quality classification prediction, with categorical boosting (CATBoost), a gradient boosting algorithm, serving as the ensemble meta-classifier and achieving the highest accuracy [46]. Based on four fundamental machine learning classification models, a study has designed a

hard voting classifier model, reaching the highest predictive accuracy of 99.5% through this ensemble learning approach [47].

Although ensemble learning has demonstrated potential in water quality prediction, there is room for optimization in the existing ensemble methods and strategies: the integration process often does not consider the variability in predictions from different models. This variability can stem from differences in model focus, specific assumptions and limitations of the models, and their varying performance. Additionally, we aim to emphasize the direct relationship between water quality parameters and categories, avoiding the reliance on complex and volatile WQI calculations. Previous studies often simply select the best-performing model, which is not a comprehensive approach to rule learning as they have not been tested against water quality determination standards; thus, deploying machine learning models without proper validation can lead to issues.

This study aims to explore how to optimize ensemble learning methods to improve the accuracy and stability of water quality predictions. Individual classification models exhibit errors and different areas of focus, often paying particular attention to the variability of certain water quality data. It is necessary to synthesize these perspectives. Variability in water quality classification is often related to parameter concentrations, such as the permanganate index, total phosphorus concentration, or WQI value, which correlate with poorer water quality categories. The discrete mapping between water quality parameters and categories is well-suited to the modeling philosophy of decision trees. Therefore, this paper designs an ensemble strategy combining ensemble tree models and voting algorithms. The ensemble tree model synthesizes individual decision tree classifiers, while the voting algorithm amalgamates the ensemble tree models. Hard voting leverages inter-model diversity to reduce overfitting common in individual models. Soft voting considers differences in model predictive performance and additional information like prediction probabilities, allowing for the integration of metrics such as accuracy, recall, precision, and F1 score to rank and weigh classifiers accordingly.

Importantly, this study does not merely filter out the best model; it considers scenarios where algorithms may err, as it is crucial to verify if the model has truly learned the rules governing water quality features. This approach is accounted for by combining the two ensemble mechanisms. The results show that the voting algorithm's robustness in correctly predicting the majority of misclassified data exemplifies its efficacy, demonstrating the dual ensemble mechanism's ability in water quality data monitoring. The water quality prediction and monitoring methods discussed in this paper were initially applied to a prefecture-level city and subsequently extended to all water monitoring stations in Beijing, where the model consistently exerted its intended functions. This provides a foundation for future applications and expansions to similar monitoring stations nationwide. We also hope to offer avenues and support for both macro and micro water environment monitoring globally.

Finally, the rest of the research work is as follows. Section 2 involves data preparation and preprocessing, as well as problem description. It also introduces ensemble tree models and voting algorithms for water quality prediction. Section 3 is the comparison of the results and model performance analysis and discusses the contribution of the model to water quality data supervision. Section 4 is about result evaluation and future work.

2. Materials and Methods

2.1. Data Preparation and Preprocessing

Water quality is intricately linked to the quality of human living environments. Pristine water quality is beneficial to sustainable societal development. This paper sourced water quality data from the China National Environmental Monitoring Centre (CNEMC), available at https://www.cnemc.cn/sssj/, 24 December 2023, focusing on the city of Qingdao. Due to the constraints of real-time data publications, the paper obtained a total of 29,448 water quality monitoring data points published by the monitoring center from 1 May 2020, at 4:00:00, to 19 December 2023, at 16:00:00. There were five monitoring sta-

tions in total: Jili River Reservoir outlet, Mo River estuary, Laoshan Reservoir center, XieLa Bridge, and Malianzhuang. The frequency of data release for the monitoring stations was once every four hours, enabling real-time water quality monitoring. In the original dataset, some of the data concerning water quality categories or factors affecting water quality were missing, including instances of consecutive time periods of missing data. Station status was categorized into two states, normal and maintenance, with maintenance indicating the aforementioned missing data situations. In cases where the sample size was large, individual samples containing missing values were directly deleted. Afterwards, the effective number of samples was 28,478.

The raw data on water quality were classified into six major categories: Category I, Category II, Category III, Category IV, Category V, and inferior Category V. According to the "Surface Water Environmental Quality Standards (GB3838-2002)" [48] (hereinafter referred to as Standard GB3838), the classification work at Qingdao's monitoring stations is conducted manually, supplemented by costly detection equipment and laboratory conditions. However, the basis for distinguishing between Category V and inferior Category V is not explicitly defined in CNEMC, and both are considered the worst categories of water quality. For the purposes of water quality monitoring and control, environmental protection personnel treat these two categories similarly in terms of attitude and resolution methods. Meanwhile, considering the issue of multi-class imbalance in machine learning, water quality prediction is influenced by the quality of the raw data and should focus on the data themselves rather than on over-sampling or under-sampling techniques. Two sampling methods disrupt the mapping relationship between water quality and feature variables. Tree models, which iteratively select the optimal features, will learn this deteriorated relationship, making it detrimental for the accurate detection of real-world water quality. Therefore, Category V and inferior Category V were regarded as equally significant and were designated as Category 4, indicating the poorest water quality. To enhance the tree model's recognition and response to water quality outcomes, we have encoded water quality into five major categories: Category 0 corresponds to Category I (best water quality), Category 1 corresponds to Category II (good water quality), Category 2 also corresponds to Category II (moderate water quality), Category 3 corresponds to Category III (poor water quality), and Category 4 corresponds to Category V and inferior Category V. As shown in Table 1, this encoding resulted in a right-skewed distribution, with an optimization of the category distribution as much as possible.

Table 1. Water quality category distribution.

Water Quality Category	Number of Instances
0	1167
1	10,060
2	8152
3	6703
4	2396

Considering the impact of physical factors and concentrations of chemical substances, there are nine control factors for water quality categories which include temperature, pH, dissolved oxygen, permanganate index, ammonia nitrogen, total phosphorus, total nitrogen, conductivity, and turbidity. Descriptive statistics of the data are shown in Table 2.

Throughout the year, the water temperature fluctuates but remains above zero degrees across all monitoring stations, with the lowest reaching nearly 0 °C. The pH generally stays within the range of 6 to 9, although occasionally peaking close to 10, indicating a tendency towards alkalinity. Dissolved oxygen concentration and permanganate index exhibit considerable variability, warranting close attention to their temporal fluctuations. Total nitrogen, total phosphorus, and ammonia nitrogen, as primary indicators of water eutrophication, demonstrate distinct patterns: total nitrogen concentrations display significant variance, total phosphorus concentrations show less variability, and all three

indicators have mean values exceeding their medians. Skewness values of 5.9911, 8.8076, and 1.7594, respectively, indicate a positive skew influenced by outliers with larger absolute values. Similarly, conductivity and turbidity exhibit substantial dispersion, with skewness values of 0.9868 and 9.3544, respectively, indicating positive skewness in their distributions. The quartile differences, particularly between Q3 and the maximum values, are more pronounced compared to the minimum and Q1. In summary, this right-skewed distribution of water quality influencing factors underscores the importance of water quality supervision and control.

Table 2. Water quality impact variables.

Statistic	Water Temperature (°C)	pH	Dissolved Oxygen (mg/L)	Permanganate Index (mg/L)	Ammonia Nitrogen (mg/L)
Mean	16.4611	8.0620	10.0397	4.6804	0.1261
Std	8.7491	0.4294	3.5515	2.4769	0.3148
Min	0.0160	6.0739	0.0040	0.2500	0.0110
Q1	8.6210	7.7600	7.5823	2.7950	0.0250
Median	16.9000	8.0550	10.0690	4.1208	0.0340
Q3	24.4800	8.3450	12.2530	6.0252	0.0716
Max	34.6520	9.8260	29.9630	46.7642	4.8410
Statistic	**Total Phosphorus (mg/L)**	**Total Nitrogen (mg/L)**	**Conductivity (μ·S/cm)**	**Turbidity (NTU)**	
Mean	0.0631	6.6932	943.6968	9.1244	
Std	0.0848	7.4637	702.3897	14.1011	
Min	0.0050	0.0500	0.2920	0.0540	
Q1	0.0171	1.8070	362.0840	2.7960	
Median	0.0288	2.9580	911.1750	4.8130	
Q3	0.0709	9.4560	1206.7903	9.1200	
Max	4.1820	34.6050	3243.3100	500.0000	

2.2. Ensemble Trees and Voting Models

Utilizing five tree models, RF, XGBoost, AdaBoost, light gradient boosting machine (LightGBM), and gradient boosting decision tree (GBDT), we embrace the concept of ensemble learning to predict water quality categories. Unlike traditional models such as KNNs or SVMs, which output a single prediction after training, ensemble methods consider multiple aspects, such as employing several decision trees, training across different iterations, and accounting for errors. The dataset was partitioned and subsequent modeling predictions were conducted using the Python programming language. Eighty percent of the data were used as the training set, while the remaining twenty percent was used as the test set. The Python version used was 3.11.8. For all ensemble tree models, we employed 4-fold cross-validation on an 80% proportion of the training dataset. Cross-validation reduces the random dependency on specific test sets, as seen in the holdout method, by partitioning the training set into multiple subsets and performing training and validation on each subset. This approach leads to a more comprehensive and reliable assessment of model performance. Parameter tuning was conducted using randomized grid search. In the process of hyperparameter tuning, we also introduced a multi-step iterated optimization strategy. We considered multi-step cross-validation, training the model multiple times in cycles and re-optimizing hyperparameters in each cycle. This approach aims to explore possible hyperparameter combinations more thoroughly, increasing the probability of finding optimal hyperparameters. Randomly selected parameters and performance scores are recorded after each training session, with higher-scoring parameter sets replacing previous ones. This method circumvents the time-consuming nature of grid search while multiple trainings help avoid the entrapment of parameters in local optima. The metric for parameter selection evaluation was accuracy. Independent test set validation is the

final prediction process, with specific model prediction performances illustrated in the confusion matrices.

Three common frameworks in ensemble learning are bagging, stacking, and boosting. The tree models mentioned above incorporate these ensemble learning concepts. Moreover, there are strategies to combine learners, such as averaging, voting, or stacking. Among these, voting involves aggregating the predictions from multiple base classifiers and selecting the most voted category or considering the highest-class probability as the final prediction. The classification prediction of water quality is further refined by employing both hard and soft voting techniques with tree models, showcasing a multi-level ensemble application.

2.2.1. Random Forest

RF is an ensemble learning method, also known as the bagging method or bootstrap aggregating. It involves sampling with replacement from the training set to form subsets, which serve as the training sets for each base model. Predictions from trained base models are then aggregated to produce the final prediction. RF utilizes decision trees as base models, randomly sampling from the training set and selecting subsets of features for training each decision tree. Typically, classification and regression trees (CARTs) are used, capable of handling both classification and regression problems. CARTs can directly handle numerical data and utilizes binary splitting. After training multiple decision trees, their predictions are aggregated through voting to arrive at the final prediction.

Key parameters determined through random search for the RF algorithm are presented in Table 3.

Table 3. RF algorithm parameter selection.

Parameter Name	Parameter Value
random_state	24
n_estimators	25
min_samples_split	3
max_features	0.8
max_depth	12
criterion	Gini

The results of many machine learning methods can be overly dependent on the influence of the set random seed [49], with random seeds causing variations in model performance [50]. It is essential to ensure the reproducibility of machine learning models and mitigate the impact of random seeds [50,51]. The parameter random_state represents the random seed, and for the five ensemble tree models, the random seed is set to 24 to facilitate reproducibility of results and reduce the impact of the seed on model stability [52]. This configuration also aims to enhance the model's focus on learning features, thereby reducing the impact of variability in subsample selection on the performance of all ensemble tree models. The n_estimators parameter denotes the number of decision trees used; here, 25 decision trees are selected as the base classifiers for the random forest. Since a single decision tree can classify an unknown sample into its predicted category, the diversity among decision trees (due to different feature selections) and their ability to vote lead to more accurate predictions than a single decision tree alone. The min_samples_split parameter refers to the minimum number of samples required at an internal node for further splitting, essentially the minimum number of samples for leaf node splitting. A value of 3 is chosen here to prevent the model from learning flawed small local samples, which could otherwise lead to overly complex trees and reduced model accuracy. The max_features parameter specifies the maximum proportion of features each decision tree uses, set to a maximum of 80% of the training set features instead of all, ensuring diversity among different trees. This is crucial in water quality classification as different features hold varying significance; for instance, the permanganate index may be important for certain

rivers, while total phosphorus concentration could be critical for lakes and reservoirs. The max_depth parameter limits the maximum depth of the decision trees to 12; the depth of a tree is the number of nodes along the longest path from the root to the furthest leaf. Together with min_samples_split, it helps control tree size and prevent overfitting. Finally, the criterion parameter refers to the metric for selecting and splitting the most important features, with the Gini impurity chosen as the criterion. For nodes containing multiple classes, it assesses the proportion of each class within the node and is a decreasing function of that proportion; smaller values make it easier for a feature to be selected by the decision tree.

2.2.2. Extreme Gradient Boosting

XGBoost, short for extreme gradient boosting, constitutes an advanced boosting technique distinct from the bagging approach, which involves parallel training of various classifiers. Instead, XGBoost sequentially trains models with subsequent models focusing on correcting the errors of the preceding one. This method enhances precision by iteratively concentrating on samples misclassified by previous models, gradually yielding classifiers capable of fine-tuning predictions cumulatively. Contrary to the equal weighting of classifiers in bagging, boosting assigns variable weights based on classifier accuracy. XGBoost is primarily composed of CARTs, facilitating the reuse of features, with each tree learning the residuals between the target value and the sum of predictions from all prior trees. It is also adaptable to other types of weak learners.

The efficiency of XGBoost hinges on its objective function, which blends the loss part—measuring model fit through the discrepancy between predicted and actual values—with a regularization term (costing tree leaf node numbers and scores) to mitigate overfitting. This dual-component objective function is optimized through gradient descent, aiming to minimize loss while controlling model complexity. Incorporating the second-order Taylor series of the tree model into the objective function allows for a step-by-step optimization process, culminating in a function concerning model parameters, including leaf count and weight.

The optimization process involves calculating the partial objective function values for samples within each leaf and aggregating these across all leaves to ascertain the overall objective function. This includes determining the optimal weight for leaves via derivation, thus serving as a metric for assessing tree model quality. Ideally, scoring all potential tree models to select the optimum is feasible, but due to the vast number of candidate models and the exigent computational demands, XGBoost employs a greedy algorithm. This algorithm optimizes by selecting features and splitting points that maximize the reduction in the objective function, iteratively splitting nodes to construct the tree model.

The training process incrementally adds newly trained models to the ensemble, progressively building the XGBoost ensemble model. This advanced approach, combining iterative error correction, model complexity control, and optimization strategies, aims to enhance predictive performance while avoiding overfitting, signifying a refined treatment of ensemble tree models by the XGBoost method.

The key parameters determined through random search for the XGBoost algorithm are presented in Table 4.

Setting the random seed to 24 serves the same purpose as in random forests, allowing the model to reproduce its feature or sample selection process rather than relying solely on randomness. The n_estimators parameter specifies the use of 24 decision trees as base models, forming a sequence where each tree attempts to correct the errors learned by the previous one. Using an appropriate number of decision trees often enhances the overall performance of the model. The min_child_weight denotes the minimum sum of the second-order gradient (weights) needed in the leaf nodes of the trees; splitting is halted if the sum is less than 1, which helps prevent the model from learning the peculiarities of local samples and thus avoids overfitting. Due to variations among samples, the subsample parameter indicates the proportion of training data used for training each decision tree;

here, a random sample proportion of 80% is used to help the model learn from higher quality samples. The maximum depth of the decision trees, max_depth, is set to 8, similar to its role in random forests, managing the complexity of the trees. The objective specifies the target function, which in this case is for multi-class classification, with multi: softmax indicating that the model outputs the class to which a sample belongs directly, rather than predicting probabilities. In gradient boosting models, loss often signifies the discrepancy between the predicted and actual values of the samples, and gamma indicates the minimum reduction in loss required to create new branches in the leaf nodes; a reduction of at least 1.2 is required for the creation of the next tree node, which also helps prevent overfitting. Multiple decision trees form an ensemble for the voting on water quality classification, and eta, the learning rate, is set at 0.3. This parameter dictates the weight of new trees added to the existing ensemble, which is done to limit the impact of each tree—particularly those with many misclassifications—on the model, allowing the optimization process to progress gradually toward the optimum.

Table 4. XGBoost algorithm parameter selection.

Parameter Name	Parameter Value
random_state	24
n_estimators	24
min_child_weight	1
subsample	0.8
max_depth	8
objective	multi:softmax
gamma	1.2
eta	0.3

2.2.3. Adaptive Boosting

AdaBoost employs the boosting concept, like XGBoost, typically using decision trees as base models. It can also utilize alternative classifiers. Each sample in the training set is assigned a weight. After initial training with equal weights, weights are adjusted based on error rates, with misclassified samples receiving higher weights. Training continues until the error rate is zero or a specified number of base models is reached. During prediction, each base model is weighted inversely to its error rate, with highly accurate classifiers having a greater influence on the final result.

The key parameters determined through random search for the AdaBoost algorithm are presented in Table 5.

Table 5. AdaBoost algorithm parameter selection.

Parameter Name	Parameter Value
random_state	24
n_estimators	25
algorithm	SAMME
learning_rate	0.55
estimator__max_depth	8

The base model comprises decision trees, with n_estimators set to 25, indicating that the model sequentially selects 25 trees to enhance learning capacity. The random_state, set to 24, ensures consistency and reproducibility across algorithms. The algorithm parameter pertains to AdaBoost's classification mechanism, where each decision tree is weighted for water quality classification. The SAMME parameter employs the error rate function for assigning weights to weak learners; trees with higher error rates receive lower weights, thus improving overall prediction by reducing the influence of poorly performing trees. Sample weights are updated using an exponential loss function, where misclassified samples in

one iteration have increased weights in the next, and correctly classified samples have decreased weights. The learning_rate, set at 0.55, controls each decision tree's contribution to the final prediction, analogous to its role in XGBoost. Additionally, the maximum depth of decision trees has been optimized to 8, balancing model complexity and performance.

2.2.4. Light Gradient Boosting Machine

LightGBM, as a lightweight gradient boosting machine, signifies an important advancement in the field of ensemble learning with regards to enhancing both efficiency and performance. Compared to its predecessor, XGBoost, this method introduces several key technological innovations during the tree-building and training processes, aimed at optimizing the utilization of computational resources, increasing the speed of model training, while striving to maintain or enhance the predictive capabilities of the model.

Firstly, LightGBM employs a histogram-based optimization algorithm that bins continuous feature values, effectively discretizing them, which significantly reduces the number of feature splitting points, thereby lowering the complexity and computational cost of model training. Moreover, this method facilitates rapid computation of gradient gains for splitting points by directly accumulating gradients within bins. This not only accelerates the search for the optimal splitting point but also induces a regularization effect through precision reduction, aiding in the prevention of model overfitting.

Secondly, the introduction of the gradient-based one-side sampling (GOSS) algorithm further enhances data processing efficiency. By prioritizing samples with larger gradients and randomly sampling those with smaller gradients, GOSS manages to reduce the volume of data processed while ensuring consistency in the data distribution, thus effectively decreasing model training time with minimal impact on model accuracy.

Thirdly, the exclusive feature bundling (EFB) algorithm addresses the issue of handling high-dimensional sparse data in model training. By merging mutually exclusive or nearly exclusive features, the EFB algorithm effectively reduces the dimensionality of the feature space, lowers the computational complexity of model training, and maintains the integrity of feature information as much as possible.

Lastly, LightGBM native support for categorical features eliminates the need for prior one-hot encoding when dealing with categorical data, thus avoiding the dimensionality explosion problem and allowing the model to more effectively utilize category features for learning.

Overall, while LightGBM bases several optimizations and innovations on the XGBoost architecture, the efficacy of its performance is intricately tied to the specific parameter configurations and characteristics of the dataset and should be selected according to actual application scenarios and requirements.

The key parameters determined through random search for the LightGBM algorithm are presented in Table 6.

Table 6. LightGBM algorithm parameter selection.

Parameter Name	Parameter Value
random_state	24
n_estimators	234
subsample	0.8
reg_alpha	0.1
reg_lambda	0.01
num_leaves	65
min_child_samples	15
max_depth	10
learning_rate	0.25

The random_state is consistently set to 24 across models to ensure reproducibility. For the base model, a notably higher number of decision trees, 234, has been selected,

likely influenced by the parameter search range and driven by accuracy metrics. This choice, moderated by additional parameters, aims to prevent overfitting. A subsample rate of 80% mitigates the influence of lower-quality samples on model learning. The parameters reg_alpha and reg_lambda represent the L1 and L2 regularization parameters, respectively. The former adds the sum of the absolute values of the model parameters as a penalty term in the objective function, often leading to sparse weights during the learning process. This encourages the model to select important features by compressing the weights of less important features to zero. The latter introduces a penalty term equal to the sum of the squared model parameters, promoting an even distribution of parameter weights and considering all relevant features, thereby enhancing the model's robustness. The values for these parameters are set to 0.1 and 0.01, respectively, ensuring both model simplification and stability. The num_leaves parameter, set at 65, governs the maximum number of leaves per tree, appropriately constrained by the dataset size for 5-category water quality classification. Its joint optimization with regularization terms enhances the model's fitting capacity. The min_child_samples parameter, akin to min_samples_split in random forests, ensures an adequate number of samples for tree splits, set at 15. Along with a max_depth of 10, these settings manage overfitting risks pre-emptively. A learning rate of 0.25 moderates each tree's impact on the final outcomes, ensuring a gradual and controlled optimization process.

2.2.5. Gradient Boosting Decision Tree

Gradient boosting decision trees utilize CARTs decision trees as their sole base model. In its objective function, GBDT includes only a loss function without a regularization component. Additionally, the error component of the loss function concerning predicted and true values is expanded only to the first-order Taylor series, with other aspects being nearly identical to those of XGBoost. However, the superior performance of either model is not predetermined, as it is influenced by multiple factors.

The key parameters identified through random search for the GBDT algorithm are presented in Table 7.

Table 7. GBDT algorithm parameter selection.

Parameter Name	Parameter Value
random_state	24
n_estimators	33
subsample	0.7
min_samples_split	2
min_samples_leaf	15
max_leaf_nodes	13
max_depth	8
learning_rate	0.18

Setting random_state to 24 ensures model consistency. The n_estimators parameter is set to 33, indicating the use of 33 trees. A subsample rate of 70% reflects the preference for using a subset of data for each tree, consistent with other tree-based models. The min_samples_split parameter, set to 2, imposes minimal restrictions on tree growth, which may reduce the model's generalization ability. The min_samples_leaf parameter, set to less than 15, ensures that each leaf node contains a sufficient number of samples, preventing the model from learning from small, flawed local samples. This also compensates for the minimal restriction imposed by the min_samples_split parameter. As a form of pre-pruning, the parameter max_leaf_nodes influences the depth and shape of the tree by limiting the maximum number of leaf nodes. Once a decision tree reaches 13 leaf nodes, it will no longer split further. This parameter functions similarly to num_leaves in LightGBM. The max_depth and learning_rate parameters, set at 8 and 0.18, respectively, ensure a

balanced influence of each tree on the final prediction outcome, preventing overfitting while maintaining model performance.

2.2.6. Hard Voting

In ensemble learning, the voting method is a technique utilized for aggregating predictions from diverse classification models. Each base model provides a predicted class label for a given input data point. Hard voting aggregates class labels predicted by base learners and selects the most voted class as the final prediction. It does not rely on classifiers to estimate probabilities, allowing application to any classifier.

2.2.7. Soft Voting

Soft voting, in contrast, considers not only the predicted classes but also the predicted probabilities for each class, aggregating models' class probabilities and selecting the class with the highest average probability as the final prediction. It permits assigning different weights to base learners, reflecting varying levels of trust based on performance metrics like accuracy, precision, recall, and F1 score. Different classification models often receive varying degrees of trust in a soft voting algorithm, which can integrate evaluation metrics to form a ranking. Based on this ranking, different levels of trust are assigned accordingly. If there are n well-ranked classifiers, adaptive weighting, exponential weighting, or a form of $n, n-1, n-2, \ldots, 1$ weighting can be applied. From the discussion below, it is evident that the performance differences among multiple ensemble tree models are not significant. However, to appropriately highlight the better-performing models, this paper employed the latter approach.

Soft voting typically outperforms hard voting by utilizing additional information—prediction probabilities—though it cannot be applied to classifiers that do not provide probability estimates. Different models often exhibit distinct attention mechanisms and learning characteristics. When encountering the same data source, the patterns of features they extract may vary. Voting algorithms can consolidate the advantages of each model as much as possible.

2.3. Model Performance Evaluation

To gain a more detailed understanding of the evaluation metrics for classification models, we present a binary classification example to help readers grasp the meaning of these metrics. In this example, class 1 represents the positive class and class 0 represents the negative class. The outputs and true results can form a confusion matrix, as shown in Table 8.

Table 8. Binary classification confusion matrix.

		Predicted Category	
		1	**0**
Actual Category	1	True Positive (TP)	False Negative (FN)
	0	False Positive (FP)	True Negative (TN)

In Table 8, TP represents the number of positive samples correctly predicted as positive, TNrepresents the number of negative samples correctly predicted as negative, FP represents the number of negative samples incorrectly predicted as positive, and FN represents the number of positive samples incorrectly predicted as negative.

The previous definitions of TP and FN, among others, do not apply to our water quality multi-classification problem. In a five-class water quality issue, each category has its own corresponding TP, TN, FN, and FP. This differs from the binary classification scenario where these are relative concepts. For instance, for Category 2, TP refers to the number of samples that are correctly classified by the model as belonging to Category 2. TN refers to the number of samples correctly classified as not belonging to Category 2, meaning it includes samples correctly identified as belonging to Categories 0, 1, 3, or 4. FP denotes the

number of samples that do not actually belong to Category 2 but are incorrectly classified as such by the model. For example, if a sample actually belongs to Category 1 but the classifier mistakenly assigns it to Category 2, it counts as an FP. In the five-class problem, concerning Category 2, FN refers to samples that actually belong to Category 2 but are incorrectly classified into Categories 0, 1, 3, or 4 by the classifier. This constitutes the performance metrics concerning Category 2 in relation to other categories. The same inferential approach can be applied to TP, TN, FN, and FP for each category, allowing us to calculate metrics such as accuracy, recall, and precision for each category. By synthesizing these metrics across all categories, we derive the overall performance metrics of the classifier. This one-to-rest analysis is solely for model metric calculations, whereas in our modeling and prediction processes, we naturally utilize the raw data.

For multi-classification problems, the cardinal purpose of model evaluation is to ensure correct classification while minimizing misclassifications across all categories. This process is then repeated for each class. Therefore, metrics such as accuracy, precision, recall, F1 score, MCC, area under the curve (AUC) value, and receiver operating characteristic (ROC) curve are highly representative and are briefly introduced as follows:

Accuracy is the ratio of correctly predicted samples to the total samples, given by Formula (1):

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FN} + \text{FP}} \tag{1}$$

Precision, also known as the positive predictive value, is the ratio of positive samples correctly predicted to the total predicted positives, calculated as (2):

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

Using this metric helps us understand the degree to which the model misclassifies, but it does not consider the magnitude of FN, which could mean missing out on reporting some positive samples.

Recall, also known as sensitivity, is the ratio of positive samples correctly predicted to the total actual positives, calculated as (3):

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

This index helps in understanding the extent of under-reporting by the model. However, it might also indicate possible over-reporting of positive samples since it does not account for the size of FP.

By combining the recall and precision metrics, one can gain a more comprehensive understanding. There are also other measures, such as the F1 score, which considers both precision and recall:

$$\text{F1} - \text{Score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{4}$$

The Matthews Correlation Coefficient, takes into account the values of TP, TN, FP, and FN, providing a comprehensive reflection of the classifier's performance. It is particularly suitable for scenarios where all aspects of classification performance need to be considered, especially excelling on imbalanced datasets. The formula is shown in Formula (5):

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \tag{5}$$

The threshold is a critical value used to convert the predicted probabilities of samples into classification labels. Generally, samples with predicted probabilities greater than or equal to the threshold are classified as positive, while those with probabilities below the

threshold classified as negative. Ideally, there exists a threshold in the trained model where samples below it are predicted as negative and are indeed negative, and samples above it are predicted as positive and are indeed positive. The ROC curve describes the degree of this ideal state. Its horizontal axis represents the false positive rate, and its vertical axis represents the true positive rate. It first sorts the predicted probabilities of the samples, continuously changes the threshold, and predicts each sample as positive. Consequently, the true positive rate and false positive rate will continuously change, thereby depicting the ROC curve. In an ideal scenario, if samples with high predicted probabilities are indeed positive, the true positive rate will continuously increase while the false positive rate remains zero. Conversely, if samples with low predicted probabilities are indeed negative, the true positive rate remains unchanged, and the false positive rate continuously increases. The ROC curve in this ideal scenario would pass through the top-left corner point (0, 1). While this ideal situation is rare, the closer the ROC curve moves towards the upper right corner, the better the model's classification performance. The closer the curve is to the upper right corner, the nearer the AUC is to 1. The AUC value, derived from the area under the ROC curve from (0, 0) to (1, 1), ranges from 0.5 to 1. Evidently, the closer the AUC value is to 1, the better the model is at distinguishing between positive and negative samples.

3. Results and Discussion

3.1. Ensemble Tree Models

The accuracy, precision, recall, and F1 score of each model are shown in Table 9.

Table 9. Ensemble tree models performance metrics.

Model	Accuracy	Precision	Recall	F1 Score	MCC
RF	0.9965	0.9958	0.9948	0.9953	0.9952
XGBoost	0.9879	0.9846	0.9853	0.9850	0.9829
AdaBoost	0.9965	0.9955	0.9947	0.9951	0.9952
LightGBM	0.9917	0.9903	0.9880	0.9891	0.9887
GBDT	0.9963	0.9963	0.9944	0.9953	0.9949

Among the five major ensemble tree models, RF and AdaBoost have the highest predictive accuracy, reaching 99.65%, with a negligible difference of 0.02% from GBDT. Meanwhile, RF has the highest recall rate at 99.48%, with its F1 score comparable to GBDT's score, both at 99.53%. Although XGBoost is an improvement over GBDT, and LightGBM is an improvement over XGBoost, GBDT exhibits the best overall performance among the three in terms of the comparison across these five evaluation metrics, especially recall rate and F1 score. Certainly, the performance of the LightGBM model outperforms XGBoost across various metrics. AdaBoost scores higher than GBDT in accuracy and recall, but its performance in precision and F1 score is slightly lower. The MCC of all five ensemble tree models is over 98%, with random forest and AdaBoost performing the best at 99.52%. Overall, the performance ranking of the five ensemble tree models, from best to least good, can be as follows: RF, GBDT, AdaBoost, LightGBM, and XGBoost.

ROC curves for each model have been plotted, with the AUC values calculated. All models achieved an AUC of 1 across all categories, with the ROC curves essentially passing through point (0, 1), indicating the high discriminatory power of each model in predicting water quality classes. Nevertheless, a detailed study of the confusion matrices for each model can provide clearer insights.

Finally, regarding the impact of various features on water quality classification, the ensemble tree models offer differing opinions, as shown in Table 10.

Excluding LightGBM, all other models consider the permanganate index as the most critical feature, with GBDT, RF, and XGBoost values exceeding 0.5—GBDT even reaching 0.6213, indicating significant influence on water quality classification. In contrast, the role of this feature is not particularly prominent in AdaBoost and LightGBM. In AdaBoost, dissolved oxygen and total phosphorus concentration are considered significant factors,

while in LightGBM, total phosphorus concentration is deemed the most important. The pH value plays a role in the decision-making process of the models but is not dominant, typically ranking fourth in most models. Dissolved oxygen concentration is viewed as the second largest factor by all ensemble tree models. In assessing the three indicators of eutrophication—ammonia nitrogen, total phosphorus, and total nitrogen—total phosphorus concentration stands out in indicating water quality, whereas the influence of ammonia nitrogen concentration is only noted in AdaBoost, ranking as the fourth important factor. Total nitrogen concentration has minimal impact on water quality classification. Meanwhile, water temperature, conductivity, and turbidity have not shown a significant influence, except for turbidity receiving higher scores in LightGBM.

Table 10. Feature importance of ensemble tree models.

Feature Variables	RF	XGBoost	AdaBoost	LightGBM	GBDT
Water Temperature	0.0069	0.0070	0.0375	952	0.0008
pH	**0.0496**	**0.0726**	**0.0842**	**1117**	**0.0444**
Dissolved Oxygen	**0.2048**	**0.1765**	**0.1350**	**1360**	**0.2198**
Permanganate Index	**0.5616**	**0.5914**	**0.3614**	**1239**	**0.6213**
Ammonia Nitrogen	0.0213	0.0280	**0.0869**	764	0.0179
Total Phosphorus	**0.1255**	**0.0766**	**0.1339**	**1472**	**0.0783**
Total Nitrogen	0.0050	0.0122	0.0442	921	0.0016
Conductivity	0.0212	0.0297	**0.0529**	875	0.0156
Turbidity	0.0041	0.0059	0.0640	**1043**	0.0003

Overall, the significance of water quality features varies across models, with pH, dissolved oxygen, permanganate index, and total phosphorus identified as most critical.

3.2. Ensemble Voting Algorithms

The results of hard voting and soft voting are presented in Table 11, where only one parameter is needed, specifically the weights in the soft voting algorithm. Based on the ranking in the model evaluation, RF, GBDT, AdaBoost, LightGBM, and XGBoost, a clear idea is to assign classifier weights as [1–5].

Table 11. Voting algorithms performance metrics.

Model	Accuracy	Precision	Recall	F1 Score	MCC
Hard Voting	0.9968	0.9957	0.9953	0.9955	0.9957
Soft Voting	**0.9972**	**0.9969**	**0.9964**	**0.9966**	**0.9961**

Among the ensemble tree models, the RF model shows the highest accuracy at 99.65%, while hard and soft voting algorithms improve this by 0.03% and 0.07%, respectively. Considering other performance indicators, soft voting appears superior, especially in terms of recall, surpassing the top-performing RF model by 0.16%. Although hard voting also shows an increase in recall and F1 score, its precision is lower than both RF and GBDT. Compared to the XGBoost model, which exhibits the worst performance, the MCC value of the soft voting method shows the highest improvement, increasing by 1.32%. Comparing the two voting algorithms, soft voting demonstrates the best performance.

Finally, due to the nature of the hard voting algorithm, which does not allow for the retrieval of prediction probabilities, only the soft voting model's ROC curve and AUC value are plotted, as shown in Figure 1. The voting method incorporates predictions from the tree models, resulting in a less pronounced zig-zag at the (1, 0) position, indicating fewer misclassification occurrences. The dashed line in the figure represents the result of a classifier making random classifications, which would result in an AUC value of only 0.5, indicating that the model lacks the ability to differentiate among water quality categories.

However, only misclassifications in Category 3 were observed, necessitating a closer look at the confusion matrices for detailed analysis.

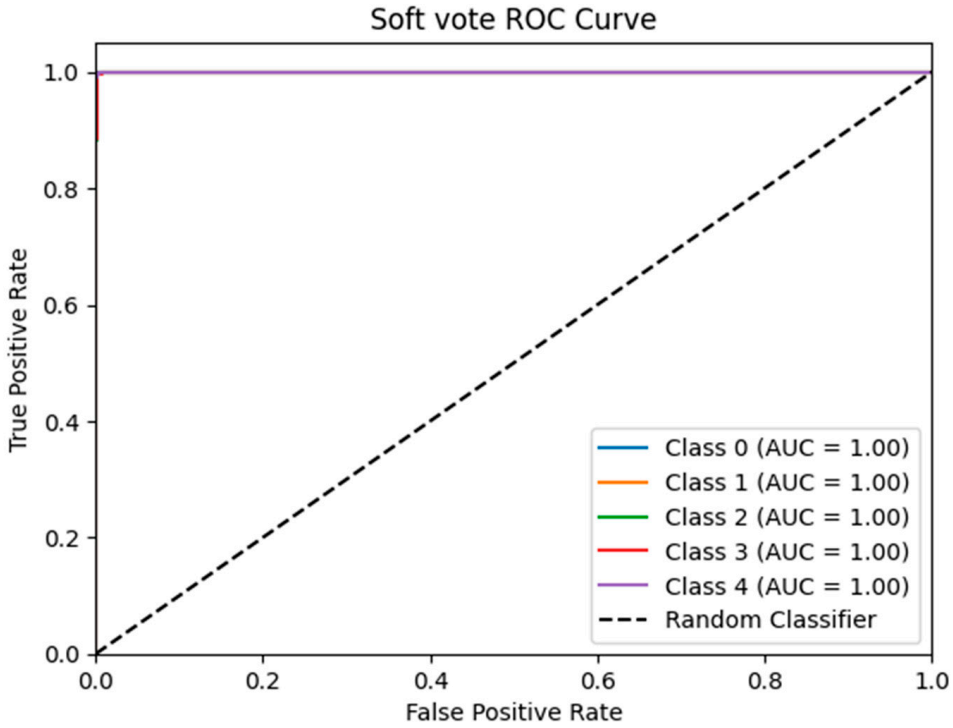


Figure 1. ROC curve and AUC value of the soft voting model.

3.3. Classifier Error Performance

Through the ensemble from tree models to the voting method, the soft voting method ranks first with a correctness rate of 99.72%. This can provide a reference and consultation for water quality monitors when determining the quality of water. In Figure 2, the confusion matrix is output to examine the error scenarios of various classification models.

Figure 2 displays the distribution of the predicted versus actual water quality categories for each classifier, related to the evaluation metrics. A sample is correctly predicted if and only if it lies on the diagonal line of the matrix. Samples located off the diagonal indicate discrepancies between the actual and predicted water quality categories, representing incorrect predictions. The best-performing soft voting method makes incorrect predictions for 16 samples, while the worst-performing XGBoost model incorrectly predicts 69 samples. Across all classifiers, there is a higher incidence of prediction errors when the actual water quality category is 3, with these often being misclassified as adjacent categories, such as 2 or 4. For the tree models LightGBM and XGBoost, which have higher error rates, there are also frequent incorrect predictions when the actual water quality category is 4. For other categories, only XGBoost demonstrates notable inaccuracy in predicting the actual Category 1.

We employed cross-validation, aligning with the common practice among most water quality prediction researchers [18,34,39,40,46]. However, ensemble tree models still exhibit bias, leading to uncertainty in sample predictions. Utilizing a secondary ensemble voting algorithm can reduce prediction errors. Furthermore, in the field of water quality monitoring, it is essential to consider more extensive model tuning parameter methods. Both nested cross-validation and out-of-sample bootstrap validation could serve as effective validation techniques, as they often result in reduced variance and bias [53].

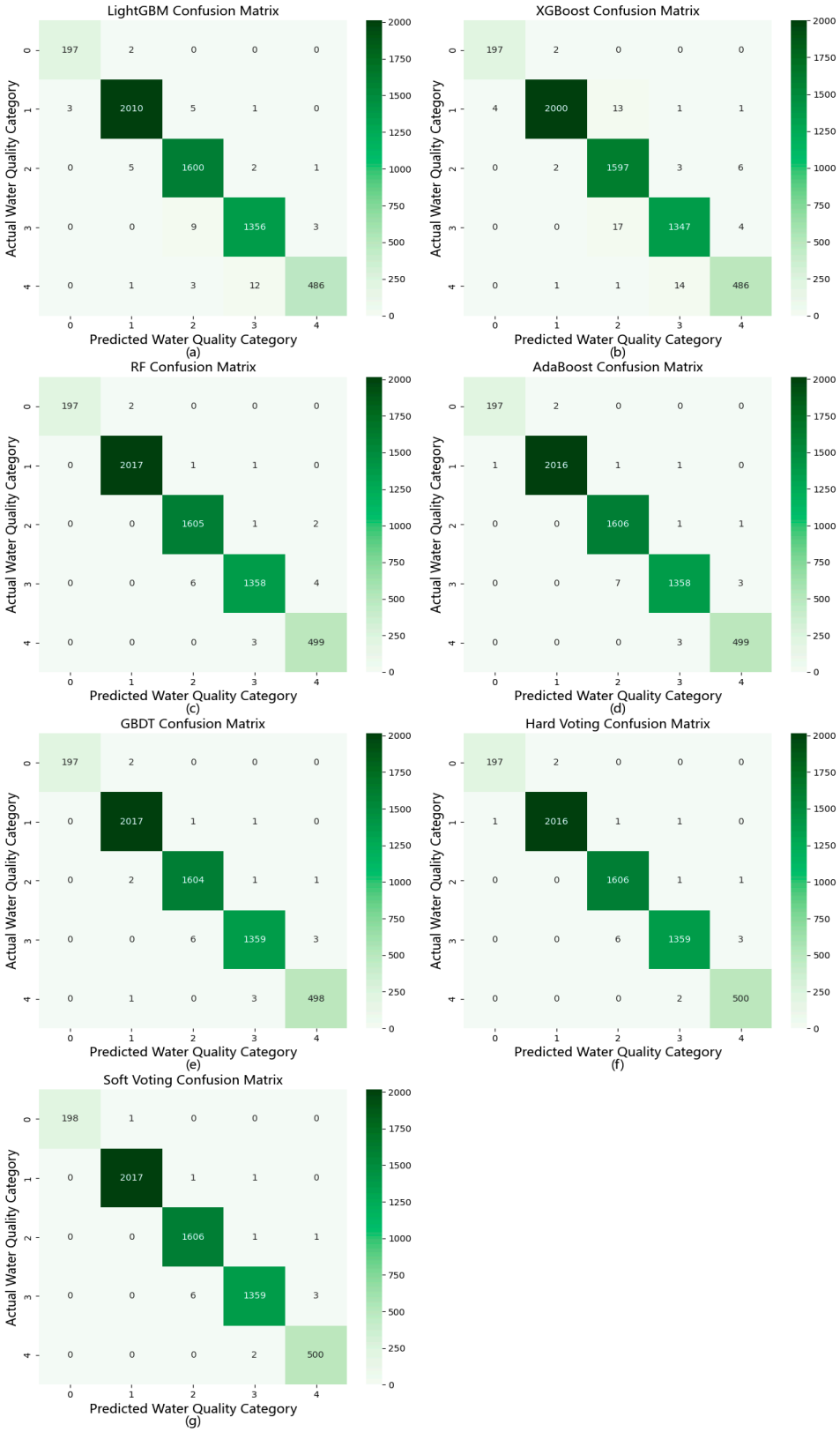


Figure 2. Error prediction scenarios of models. (**a**) Confusion matrix of LightGBM classifier, (**b**) confusion matrix of XGBoost classifier, (**c**) matrix of random forest classifier, (**d**) confusion matrix of AdaBoost classifier, (**e**) confusion matrix of GBDT classifier, (**f**) confusion matrix of hard voting classifier, and (**g**) confusion matrix of soft voting classifier.

3.4. Effectiveness Analysis of Two Ensemble Mechanisms

Firstly, a substantial amount of training work and parameter tuning was conducted prior to producing the output results, ensuring that the ensemble tree model achieved a certain level of precision. Secondly, considering the characteristics of the dataset based on the standard GB3838's classification of water quality data, the feature values and water quality category labels coexist in such a manner: features are discretized into intervals, which are then mapped to unique target labels. Decision tree models are designed to construct the tree by selecting the best features for splitting, thus enabling them to naturally handle discretized feature values and intuitively apply these intervals in the decision-making process of the model. The relationship between key features and water quality categories according to the standard GB3838 is shown in Table 12. In the second row of the total phosphorus index, the criteria in parentheses refer to the classification standard for monitoring stations in lakes and reservoirs. This classification standard is based on the single-factor evaluation method, which selects a critical pollutant indicator or environmental parameter, evaluates it by comparing its concentration with the established environmental quality standards or thresholds, and determines whether the environment exceeds the pollution limit or meets a certain quality level. Finally, this method integrates the judgment of multiple indicators, classifying water quality into the poorest category corresponding to a given parameter.

Table 12. Surface water environmental quality standards GB3838.

Feature Indicator	Class 0	Class 1	Class 2	Class 3	Class 4
pH (dimensionless)			All 6–9		
Dissolved Oxygen (mg/L)	7.5 and above	[6, 7.5)	[5, 6)	[3, 5)	Below 3
Permanganate Index (mg/L)	[0, 2]	(2, 4]	(4, 6]	(6, 10]	Above 10
Ammonia Nitrogen (mg/L)	[0, 0.15]	(0.15, 0.5]	(0.5, 1.0]	(1.0, 1.5]	Above 1.5
Total Phosphorus (mg/L)	[0, 0.02] ([0, 0.01])	(0.02, 0.1] ((0.01, 0.025])	(0.1, 0.2] ((0.025, 0.05])	(0.2, 0.3] ((0.05, 0.1])	Above 0.3 (0.1)

Next, the tree-based voting algorithms are compared with three classical machine learning algorithms: LR, KNNs, and gaussian naive bayes (GNB). Four-fold cross-validation with random search optimization was conducted for these models, and data cleaning and normalization were performed according to the characteristics of each algorithm. The classical machine learning algorithms and the tree-based voting algorithms were learned on the same dataset. The results are summarized in the following table, Table 13.

Table 13. Classic machine learning algorithms performance metrics.

Model Name	Accuracy	Precision	Recall	F1 Score
LR	0.7590	0.7772	0.7311	0.7507
KNNs	0.9133	0.9122	0.9024	0.9070
GNB	0.6986	0.6887	0.7173	0.6845

Among the three classic machine learning algorithms, the KNNs model performs the best. However, its accuracy and other metrics are still significantly lower than those of the worst-performing XGBoost model among the ensemble tree models. This indicates that the applicability of machine learning algorithms is linked to the characteristics of the dataset, and even with a high-quality dataset, models can have varying levels of performance. In the second point of discussion, we highlight the characteristics of the dataset in the Chinese water quality monitoring system, where tree models achieve the best classification results, whereas the three classic machine learning algorithms somewhat overlook local data relationships.

We are concerned that some ensemble tree models may overlook the relationship between local features and categories. Therefore, a voting algorithm is necessarily employed to summarize the specific attributes discovered by each model in the water quality

data, preventing potential prediction errors caused by variations in focus and setup across ensemble trees. In cases where large sample data are released by monitoring stations, a comparative analysis of the improvements in accuracy and precision indicators through the voting algorithm indicates that ensemble tree models only slightly neglect this relationship. In such scenarios, significantly enhancing predictions with the voting algorithm is challenging, but introducing cost-free additional information can provide slight improvements, which is facilitated by the weighting of base models and prediction probabilities in the soft voting algorithm.

The voting algorithm aggregates the predictive results of high-precision ensemble tree models. However, based on the validation of the voting algorithm, selecting only a single best ensemble tree model instead of the voting algorithm has its limitations. Observing the distribution of predicted versus actual categories in the confusion matrix reveals differences in the tree model predictions. Yet, through the voting algorithm, an interesting commonality emerges: certain data points appear to be consistently misclassified across all models. Given that tree models are well-adapted to the dataset's features, these mispredicted data points warrant careful attention. As discussed further below, simply opting for the best-performing RF model might overlook these critical insights.

3.5. Water Quality Data Supervision

The reasons for classifier prediction errors are diverse, encompassing low fitting degree and performance of the classifier itself, as well as quality issues in the original data. Employing fitted models, erroneous classifications were examined through voting. From the confusion matrix, it is evident that individual ensemble models exhibit randomness in predicting certain samples. Thus, under two types of voting algorithms, erroneous predictions from the original data points were identified, along with the corresponding misclassifications by base classifiers. The results are presented in Table 14.

Table 14. Distribution of misclassified data.

Original Class	Predicted Class	Sample Point Position	Misclassified Base Classifiers
3	2	10,904	RF, XGBoost, AdaBoost, LightGBM, GBDT
3	2	8018	RF, XGBoost, AdaBoost, LightGBM, GBDT
1	3	7016	RF, XGBoost, AdaBoost, LightGBM, GBDT
4	3	7014	RF, XGBoost, AdaBoost, LightGBM, GBDT
0	1	690	RF, XGBoost, AdaBoost, LightGBM, GBDT
2	4	26,804	RF, XGBoost, AdaBoost, LightGBM, GBDT
3	2	523	RF, XGBoost, AdaBoost, LightGBM, GBDT
3	2	519	RF, XGBoost, AdaBoost, LightGBM, GBDT
1	2	491	RF, XGBoost, AdaBoost, LightGBM, GBDT
4	3	24	RF, XGBoost, AdaBoost, LightGBM, GBDT
3	2	13,732	RF, XGBoost, AdaBoost, LightGBM
2	3	87	RF, XGBoost, AdaBoost, GBDT
3	2, 4	49	RF, XGBoost, AdaBoost, GBDT
3	4	6823	RF, AdaBoost, GBDT
3	4	12	RF, XGBoost, GBDT

In the field of environmental monitoring and water quality assessment, classifiers are extensively utilized to automatically categorize water body data from monitoring stations into predefined water quality categories. The classification criteria for monitoring stations are defined according to standard GB3838 and follow single-factor water quality evaluation methods. While this standard is clear and well-defined, the analysis below clearly indicates that the classification of original data is not always error-free.

In the feature importance rankings, the permanganate index is the most critical factor in the majority of models. However, when classification is determined based on the GB3838 standard and the voting algorithm, discrepancies are observed in the classification of the original data. The results are presented in Table 15.

Table 15. Water quality discrimination based on permanganate index samples.

Sample Number	Water Temperature (°C)	pH	Dissolved Oxygen (mg/L)	Permanganate Index (mg/L)	Ammonia Nitrogen (mg/L)	Total Phosphorus (mg/L)
12	22.5	8.35	15.21	12.5	0.22	0.23
87	21.7	8.5	12.54	9.84	0.249	0.19
491	14.1	7.78	9.34	4.03	0.025	0.023
519	13.5	7.7	6.36	5.08	0.027	0.078
10,904	4.207	8.354	14.284	5.8914	0.025	0.0286
8018	26.981	8.677	10.438	5.7319	0.0504	0.0605
49	21.7	8.16	10.16	10.95	0.307	0.229
523	16.5	8.28	13.39	6	0.039	0.15

Sample Number	Total Nitrogen (mg/L)	Conductivity (μ·S/cm)	Turbidity (NTU)	Reassessed Category by Standard GB3838	Predicted Category by Voting Algorithm	Original Assessment by Monitoring Station
12	5.87	1917.7	33.6	4	4	3
87	3.87	2001.2	46	3	3	2
491	1.85	425.9	7.4	2	2	1
519	0.63	887.9	3.7	2	2	3
10,904	4.385	1511.464	1.354	2	2	3
8018	1.332	1095.01	21.955	2	2	3
49	4.61	1984.3	37.1	4	4	3
523	8.03	2898.5	34.3	2	2	3

For sample number 49, the permanganate index is 10.95 mg/L, which should classify it as Category 4. Only the XGBoost model classified it as Category 2, while all other classifiers identified it as Category 4. The voting algorithm, along with the standard GB3838 classification, also assigned it to Category 4. There are instances where sample points fall on the classification boundaries defined by the GB3838 standard but are misclassified in the original data. For example, sample number 523 has a permanganate index that is exactly at the threshold, and both the voting algorithm and the GB3838 standard indicate that it should be classified as Category 2. Further details for other samples are provided in Table 15, where the reclassification based on the GB3838 standard aligns perfectly with the results from the voting algorithm.

One sample experienced a prediction error related to the dissolved oxygen concentration: for sample number 7016, with a dissolved oxygen concentration of 4.118 mg/L, it should be classified as Category 3, but the original data classified it as Category 1. Regarding total phosphorus concentration, sample number 7014, taken from the estuary of the Ink River, has a total phosphorus concentration of 0.263 mg/L and should be classified as Category 3; however, the original data classified it as Category 4. Similarly, for sample number 26,804, with a total phosphorus concentration of 4.182 mg/L, our classifier classified it as Category 4, consistent with GB3838.

Additionally, aside from misclassifications in the original data, there are instances where the classifiers themselves make errors: For sample number 24, with a permanganate index concentration of 14.99 mg/L, it should have been classified as Category 4, but the classifier incorrectly assigned it to Category 3. For sample number 13732, with a total phosphorus concentration of 0.2341 mg/L, it should have been classified as Category 3, yet the classifier made a mistake. For sample number 690, all data indicate it should be classified as Category 0 according to GB3838 standards, but the classifier incorrectly assigned it to Category 1. For sample number 6823, with a dissolved oxygen concentration of 3 mg/L, which is exactly on the boundary between Categories 3 and 4, it should have been classified as Category 3, but the classifier erroneously assigned it to Category 4. Only XGBoost and LightGBM classified it correctly. Therefore, it is crucial to integrate results from multiple classifiers for more accurate judgment.

In the classification of water quality monitoring data, incorrect category assignments can be attributed to various factors, including errors in the original data labels, boundary cases in category determination, and the initial performance of the classifiers. Among the 15 samples where the binary voting algorithm made errors, only 4 were due to classifier misjudgment, while the other 11 were incorrectly labeled in the original data. These labeling errors were identified through rules learned by the classifiers, and the predicted categories matched those determined by the GB3838 standard. Thus, the construction of the voting algorithm not only aids in water quality prediction but also plays a role in the oversight and control of data quality issues. This demonstrates that even when there are data quality issues, the model can effectively understand the relationships between features and categories. Particularly, when there is a discrepancy between the predicted and actual categories, it provides researchers with valuable insights into whether there might be issues with the samples themselves.

The water quality categories of the aforementioned 11 samples were re-labeled, and the performance ranking of the ensemble tree model remained unchanged, maintaining consistency with the parameters of the original training. Finally, the performance of hard voting and soft voting is shown in Table 16.

Table 16. Voting algorithms performance metrics.

Model	Accuracy	Precision	Recall	F1 Score	MCC
Hard Voting	0.9987	0.9980	0.9971	0.9975	0.9983
Soft Voting	0.9991	0.9991	0.9982	0.9986	0.9988

Selecting a single best ensemble tree model does not reveal commonalities in prediction differences due to the randomness in the base model predictions, thus limiting its effectiveness for water quality data oversight. In contrast, a voting algorithm subjected to data regulation and control achieves improved model accuracy, with the optimal model being the soft voting model, reaching a peak accuracy of 99.91%. Finally, we transferred the model for direct monitoring and testing of water quality in Beijing. The results can be found in Supplementary Materials. It must be noted that this transfer has certain limitations, as our model has not directly learned the characteristics specific to Beijing's water quality data, which could impact model performance. Nonetheless, it is evident that the dual ensemble model can capture most of the rules for water quality determination, and its adaptability is sufficient, as it effectively functions in data quality oversight.

4. Conclusions

The quality of water categories is invariably correlated with the magnitude of numerous chemical or physical factors. Historically, scholars have developed the Water Quality Index (WQI) to integrate these factors, thereby linking water quality categories through the magnitude of the WQI. This one-to-one mapping from WQI intervals or factor ranges to water quality suggests insights into constructing decision tree models. Decision tree models

are built through a series of judgements and branches, relying on identifying the optimal attribute for data partitioning. However, due to the variability in water quality data, the focus of a single tree often differs, which necessitates a comprehensive ensemble approach. We have implemented a secondary integration of voting algorithms with basic tree models, achieving an accuracy of 99.91% with soft voting. Despite the imbalance in the original data categories, our model effectively learned the mapping between the magnitudes of water quality attributes and water quality categories, with the soft voting reaching an MCC value of 99.88%.

Furthermore, this study extends beyond merely identifying the optimal model, as many machine learning processes are about learning rules, necessitating model testing for better future deployment. Our model functions as a tool for data regulation; the original data had imperfections. The voting algorithm indicated the direction of these data deficiencies, as seen in Tables 14 and 15, where the predicted categories largely aligned with those reassessed using the GB3838 standard. Finally, the direct application of the voting model under dual integration for water quality prediction in Beijing demonstrates its robust applicability.

It is crucial to recognize that our modeling data directly link parameters affecting water quality to water quality categories. This presents a complex learning task for ensemble models, yet it aptly provides an approach to the rawest water quality data without WQI calculations. Different countries have various standards for assessing water environments, which can serve as benchmarks for nations with inadequate water treatment conditions. The results suggest the necessity of aligning data modeling with its intrinsic characteristics. In many countries and regions, the magnitude of water quality parameters directly correlates with water quality categories, as illustrated in Table 12, which shows the relationship between the permanganate index or dissolved oxygen concentration and the five major water quality categories in China's monitoring areas. Thus, the concept of constructing decision tree-based models should be broadly applicable and suitable for these nations. It unveils hidden patterns in the data through a tree structure, discovering attribute conditions of multiple parameters via recursive splitting, thus aiding water quality category determination. Moreover, tree models exhibit inherent robustness against missing values and outliers, unlike WQI calculations, which often require complete datasets.

We hope our research proves practically valuable. In continuous water quality monitoring, data are variable, and in complex situations, water quality categories are not static. If tree models can effectively monitor water quality data, they remain viable. When water quality trends deviate from the monitoring standards, the algorithm can trigger warnings, and this study offers feasibility for water quality data management. Future research should explore more model diversity and different ensemble methods. Ultimately, we suggest integrating ensemble tree models with deep learning models, as both deep ensemble learning and feature learning can benefit from tree models' strengths in feature selection and handling heterogeneous data, while leveraging the power of deep learning in processing vast amounts of data and complex features.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/su162310634/s1, Table S1: Ensemble Models Performance Metrics; Table S2: Supervised Discrimination of Water Quality Data; Table S3: Water Quality Discrimination Based on Total Phosphorus Parameters.

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